

Joseph Wilson

Email: joseph.wilson@uqconnect.edu.au, Website: <https://josephwilsonmaths.github.io>

Ph: 0400234540,

Location: Brisbane, Australia

Research Summary

I am a PhD student in Mathematics at The University of Queensland, researching the theoretical foundations of modern AI systems and methods for principled uncertainty quantification in deep learning. My work combines mathematical analysis, and large-scale empirical evaluation to better understand and improve the reliability of machine learning systems.

My research interests are: Bayesian deep learning, uncertainty quantification, random matrix theory, neural tangent kernels, statistical learning theory, generalization in deep neural networks.

Education

July 2023 - Present

PhD, Mathematics - The University of Queensland

- Advisors: Professor Fred Roosta (UQ, CIRES), Dr. Liam Hodgkinson (Uni. Melb), Dr. Chris van der Heide (Uni. Melb).

2019 - 2022

Bachelor of Mathematics (Honours I) - The University of Queensland

- Honours thesis: "A Mathematical Model of Mechanical Cell Competition in Epithelial Tissue". Supervisor: Dr. Zoltan Neufeld.
- GPA 6.53, Deans Commendation for Academic Excellence, Sem. 1 2022, Sem. 2 2022, Sem. 1, 2023, Sem. 2 2023.

Publications

- *Is the Last Layer Sufficient for Uncertainty Quantification?*
Joseph Wilson, Chris van der Heide, Liam Hodgkinson, Fred Roosta
International Conference on Machine Learning (ICML), 2026, (accepted, to be published)
- *Uncertainty Quantification with the Empirical Neural Tangent Kernel*
Joseph Wilson, Chris van der Heide, Liam Hodgkinson, Fred Roosta
Conference on Neural Information Processing Systems (NeurIPS) 2025
- *Mechanical Cell Competition in a Model of Epithelial Layer with Size Dependent Proliferation*
Chenhao Zhang, Leo Clarke, **Joseph Wilson**, Faris Saad Alsubaie, Zoltan Neufeld
Mathematical Medicine and Biology: A Journal of the IMA, Volume 42, Issue 4, December 2025

Invited Talks and Presentations

- Seminar, The School of Business Analytics, The University of Sydney, NSW, AU, 2026
Is the Last Layer Sufficient for Uncertainty Quantification?
- Seminar, The University of California, Berkeley, Berkeley, CA, USA, 2025
Is the Last Layer Sufficient for Uncertainty Quantification?
- Seminar, Workshop on Optimisation, Metric Bounds, Approximation and Transversality (WOMBAT), The University of Queensland, QLD, AU, 2025
Uncertainty Quantification with the Empirical Neural Tangent Kernel
- Seminar, 5th Biennial Meeting of the Pacific Northwest Section of SIAM, The University of Washington, Seattle, WA, USA, 2025
Uncertainty Quantification with the Empirical Neural Tangent Kernel

Employment / Teaching

2022 - Present

Tutor - The University of Queensland

- Courses tutored: MATH1051, MATH1052, STAT2003, STAT2004, STAT2203, STAT3001, STAT3006, MATH3102, MATH3204.

Jan 2023 - July 2023

Control Engineer - Whitelaw Hydraulics

- Software programming, development, and on-site maintenance for control systems for agricultural and industrial applications.
- Developed AI system for self-navigation, programming languages: CodeSYS, C#.

Service

Jan 2025 - Present

Mathematical Aspects of Machine Learning (MAML) Seminar Series, UQ

Coordinator and co-founder, responsible for finding weekly speakers, notifying attendees of changes, and organizing room bookings.

Professional Development

Jan 2024 - Feb 2024

AMSI Winter School

Completed courses: *Machine Learning/Deep Learning* and *Bayesian Inference and Computations*.